

Deliverable 6: Evaluation and Expansion

Objective

Evaluate the Lung Cancer Knowledge Graph's accuracy by comparing its query results to trusted medical sources and identify opportunities to enhance its scope and coverage.

Evaluation

1. Process

The graph was evaluated by cross-verifying query results with:

- SNOMED CT: For standardized terminology.
- DrugBank: For medication data and interactions.
- MedlinePlus: For information on symptoms, treatments, and diagnostic tests.

2. Key Queries

- What are the symptoms of Lung Cancer?
- What treatments are available for Non-Small Cell Lung Cancer?
- Which medications are used in Chemotherapy?

3. Findings

- Strengths:
Accurate representation of common symptoms, treatments, and diagnostic tests.
Well-defined relationships, e.g., Lung Cancer → HasSymptom → Cough.
- Gaps:
Rare subtypes of Lung Cancer are underrepresented.
Side effects, costs, and success rates of treatments and medications are missing.
Limited integration of real-world data, such as patient reviews or clinical outcomes.

Expansion

Gaps to Address

1. Broader Disease Coverage:

Add related conditions like Asthma and Chronic Bronchitis for comparative insights.

2. Detailed Treatment Attributes:

Include side effects, success rates, and costs for treatments and medications.

3. Patient-Centric Data:

Integrate anonymized patient feedback and clinical trial results.

4. Lifestyle Interventions:

Strengthen relationships like Quit Smoking → ReducesRisk → Lung Cancer.

Proposed Data Sources

- PubMed: For updated medical research.
- WHO/CDC Databases: For global health statistics.
- NIH Clinical Trials: For real-world evidence on treatment efficacy.

Evaluation Report

Summary

1. Strengths:

The knowledge graph is accurate and well-structured for Lung Cancer-related knowledge.

Queries reliably return key insights on symptoms, treatments, and medications.

2. Recommendations:

Expand disease and treatment coverage with detailed attributes.

Incorporate real-world data to enhance practical applicability.

Conclusion

The Lung Cancer Knowledge Graph is a robust tool for healthcare insights. By addressing identified gaps, it can become a comprehensive resource for clinicians, researchers, and patients, driving better decision-making and care outcomes.